

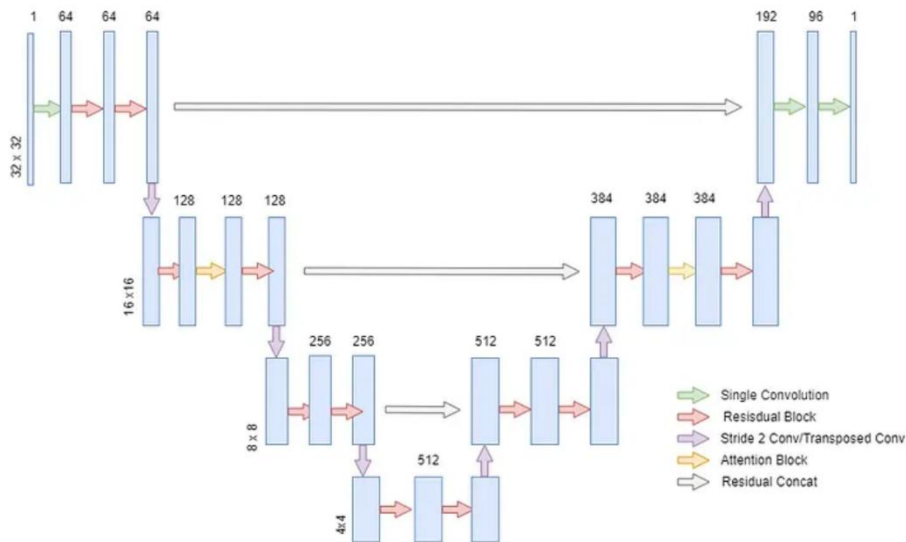
HW3: Image Generation for Handwritten Digits

1. Model Description

The image generation model implemented is a Denoising Diffusion Probabilistic Model (DDPM). The core component is a U-Net architecture that serves as the noise-prediction network. The DDPM is trained to predict the noise added to an image at different timesteps.

The U-Net employs a cascade of feature maps, beginning with a base number of channels. The channel count doubles at each down-sampling level, progressing from 96 to 192 and then to 384 channels. This progressive increase in channel capacity allows the model to capture more complex features as spatial resolution decreases. Additionally, the use of two sequential residual blocks at each level further deepens the architecture, allowing for richer nonlinear transformations at each resolution.

1.1. Architecture Illustration



1.2. Core Components & Modifications

Time Embedding: The timestep is encoded using a sinusoidal positional embedding before being processed by a two-layer Multi-Layer Perceptron (MLP) with ReLU activation. This mechanism ensures that temporal information is effectively embedded in the feature representation.

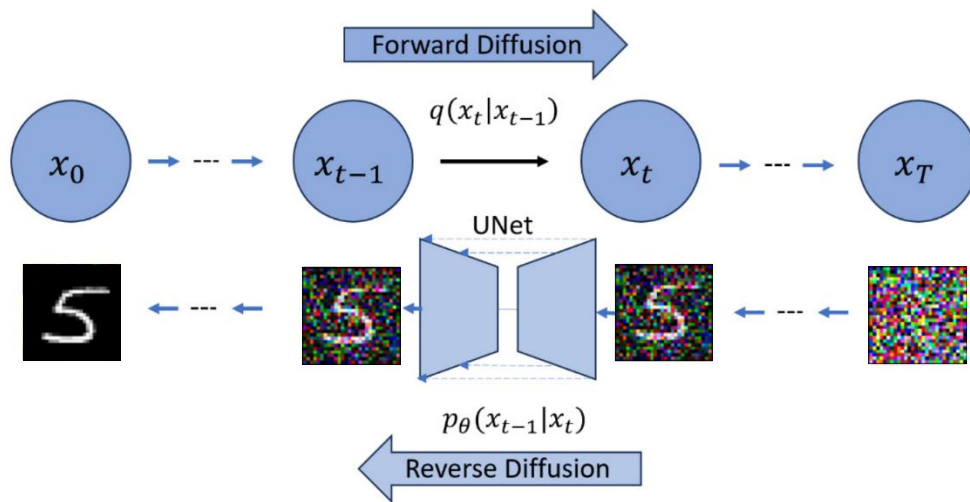
Residual Blocks: Each block consists of two convolutional layers, group normalization, a skip connection, and integrates the time embedding through an element-wise addition operation after the first convolution. The use of a skip connection allows for stable training of deep networks.

Group Normalization: Dividing feature channels into eight groups is preferred over standard Batch Normalization in diffusion models because it is more robust to smaller batch sizes and prevents information leakage between samples within a batch.

Architecture: The U-Net employs a symmetric encoder-bottleneck-decoder structure. The encoder utilizes max pooling for spatial down-sampling, and the decoder performs up-sampling via bilinear interpolation. Feature maps are concatenated with corresponding encoder outputs through skip connections, which is critical for preserving fine-grained spatial information.

Exponential Moving Average: An EMA mechanism is applied to the model parameters during training, using a decay rate of 0.9999, to stabilize learning and enhance the quality of the generated samples by averaging weights across training steps.

1.3. DDPM Training & Inference Flow



➤ Training Pipeline

Forward Diffusion: A clean image x_0 is corrupted to a noisy image x_t by sampling a random time step $t \sim \text{Uniform}(\{1, \dots, T\})$ and applying a single step of the known Gaussian noise distribution $q(x_t|x_0)$.

Noise Prediction: The U-Net (ϵ_θ) takes the noisy image x_t and the time step t as input, aiming to predict the exact noise ϵ that was added.

Loss Optimization: The model parameters θ are updated by minimizing the Mean Squared Error (MSE) between the actual noise ϵ and the predicted noise $\epsilon_\theta(x_t, t)$.

EMA Integration: After each AdamW optimization step, the EMA shadow weights are updated using a high decay rate.

➤ Inference Pipeline

Initialization: The process begins with sampling x_T from a pure Gaussian distribution, $x_T \sim N(0, I)$.

Iterative Denoising: The model iterates backward from $t = T$ down to $t = 1$. At each step, the U-Net predicts the noise $\epsilon_\theta(x_t, t)$.

Reverse Sampling: The predicted noise is used to calculate the mean of the learned reverse distribution, from which x_{t-1} is sampled. This step is governed by the learned posterior variance and is executed sequentially for all 1,000 timesteps.

Final Output: Upon completing the final iteration, the model produces the synthesized image x_0 .

2. Implementation Details

2.1. Dataset and Preprocessing

The MNIST dataset of handwritten digits was utilized, with images loaded as PNG files using a custom dataset handler. Although the original MNIST digits are grayscale, they were converted to RGB to match the U-Net model's input requirements. The image size remains 28 x 28 pixels. Images are normalized to the range [-1, 1] before being fed into the model. No explicit data augmentation was applied.

2.2. Hyperparameters

Epochs	Batch Size	Optimizer	Learning Rate	EMA Decay	Timesteps	Base Channels	Noise Schedule
300	128	AdamW	0.0002	0.9999	1000	96	Cosine

Optimizer Specification: The training employs the AdamW optimizer, recognized for its decoupled weight decay. This choice is essential for stabilizing the training of complex generative models such as DDPMs, as it alleviates the common issue of high weight decay interfering with adaptive learning rates.

Noise Schedule Specification: The default choice has shifted from the simpler linear schedule to the more effective Cosine Schedule. The linear schedule tends to assign excessive weight to large-variance steps at the beginning of the diffusion process, which can make early denoising steps overly volatile. In contrast, the Cosine Schedule, characterized by a slight offset, provides a much smoother variance decay, particularly stabilizing the later, low-noise steps.

2.3. Loss Functions

For training the noise prediction network, the Mean Squared Error (MSE) loss function is used to measure the difference between the predicted noise and the actual noise. The formula is defined as:

$$\mathcal{L} = \text{MSE}(\epsilon, \epsilon_{\theta}(x_t, t))$$

2.4. Training and Evaluation Strategies

The training strategy is focused on achieving stability and tracking convergence. Checkpoints are saved every five epochs to facilitate progress tracking and monitoring. These checkpoints include the current epoch number, model state, optimizer state, average loss, and the EMA shadow weights. This thorough checkpointing allows for robust analysis and model recovery. The inference phase involves a reverse process consisting of 1,000 steps. This sampling process is performed with the model in evaluation mode, ensuring that no gradients are

computed during image generation. The primary quantitative metric is the Fréchet Inception Distance (FID).

3. Results Analysis

3.1. Quantitative Improvements Through Multiple Experiments

Optimization was achieved by systematically tuning hyperparameters to minimize the FID score, which serves as a measure of distribution similarity between the generated and real image samples. The average loss value was also recorded on the training set during each epoch.

Experiment	Configuration	FID
Case 1	epochs: 100, batch size: 128, optimizer: AdamW, learning rate: 0.0002, base channels: 64, noise schedule: linear, residual blocks: 1 layer per level	2.77
Case 2	epochs: 150 , batch size: 128, optimizer: AdamW, learning rate: 0.0002, base channels: 64, noise schedule: linear, residual blocks: 1 layer per level, EMA (decay rate = 0.9999)	1.71
Case 3	epochs: 200 , batch size: 128, optimizer: AdamW, learning rate: 0.0002, base channels: 80 , noise schedule: linear, residual blocks: 1 layer per level, EMA (decay rate = 0.9999)	1.55
Case 4	epochs: 200, batch size: 128, optimizer: AdamW, learning rate: 0.0002, base channels: 80, noise schedule: cosine , residual blocks: 1 layer per level, EMA (decay rate = 0.9999)	1.44
Case 5	epochs: 300 , batch size: 128, optimizer: AdamW, learning rate: 0.0002, base channels: 96 , noise schedule: cosine, residual blocks: 1 layer per level, EMA (decay rate = 0.9999)	1.08
Case 6	epochs: 300, batch size: 128, optimizer: AdamW, learning rate: 0.0002, base channels: 96, noise schedule: cosine, residual blocks: 2 layers per level , EMA (decay rate = 0.9999)	1.03

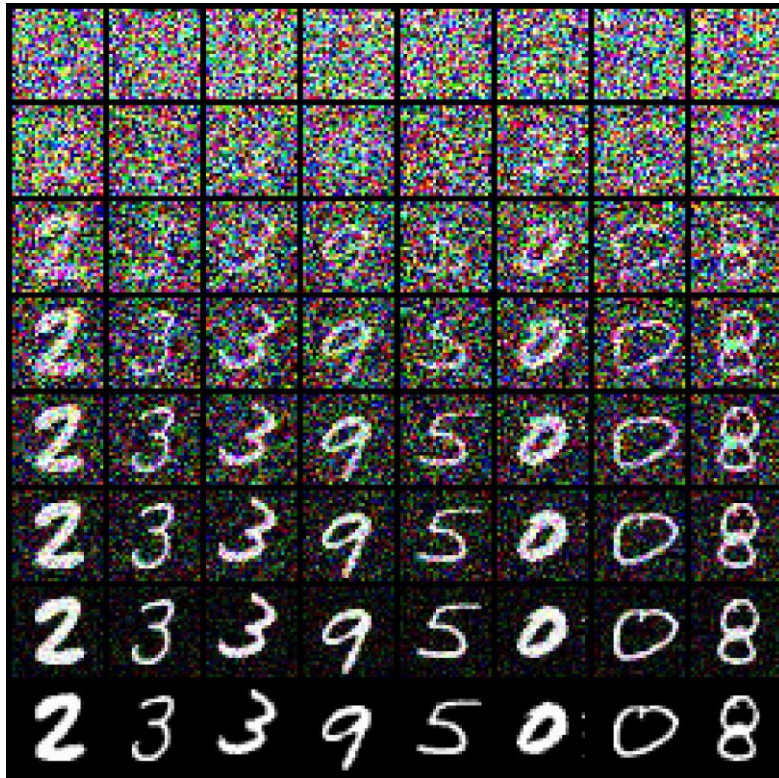
Interpretation of Results

The experiments indicate that the performance of DDPMs is strongly influenced by both model capacity and regularization. EMA acts as an effective stabilizer, reducing weight fluctuations and producing immediate improvements in FID scores. Increasing model capacity—by expanding the base channel width and prolonging the training duration—further enhances performance. Additionally, the Cosine Noise Schedule contributes gains by smoothing the variance profile across timesteps. Collectively, these factors enable the U-Net architecture to capture more precise feature relationships and model the reverse diffusion dynamics more accurately.

Increasing the number of residual blocks per level proved critical. Expanding each encoder and decoder stage from one to two sequential residual blocks enhanced the network’s nonlinear representation capacity at each spatial scale. This additional depth improved the model’s ability to extract fine-grained structural cues and

more accurately approximate the underlying noise distribution, directly contributing to the lower FID scores achieved by the higher-capacity configurations.

3.2. Diffusion Process Visualizations



4. Short Conclusion

The Denoising Diffusion Probabilistic Model, based on a high-capacity, two-layer Residual Block U-Net architecture, demonstrated superior image generation capabilities on the MNIST dataset. The optimal configuration—featuring a Cosine Noise Schedule, Exponential Moving Average regularization, and 300 training epochs—achieved a final Fréchet Inception Distance score of 1.03. This outstanding performance confirms the model's ability to accurately approximate the reverse diffusion process. Although the current sequential sampling process is computationally intensive, the generated images exhibit high perceptual quality. Future research should focus on implementing accelerated sampling methods to enhance inference efficiency without compromising the demonstrated fidelity.